

Lab 2

Project Requirements

Phase 1: Data Loading & Inspection

- Create Spark session
- Load dataset using `sc.textFile()`
- Remove header
- Split rows into columns
- Display sample records
- Count total records
- Count number of partitions

Required operations

Must use:

- `map()`
- `filter()`
- `take()`
- `getNumPartitions()`

Phase 2: Data Cleaning

Using **RDD transformations only**:

- Remove missing values
- Remove duplicate records
- Fix malformed rows
- Trim whitespace
- Convert numeric columns to proper types

Phase 3: Core RDD Transformations

must demonstrate **at least SIX** different RDD transformations/actions.

Choose from:

- map()
- flatMap()
- filter()
- distinct()
- union()
- intersection()
- subtract()
- sample()
- mapPartitions()
- coalesce()
- repartition()

For each:

- Show code
- Explain purpose
- Show output example

Phase 4: Key-Value Operations

Required operations (at least FOUR)

- reduceByKey()
- groupByKey()
- countByKey()
- sortByKey()
- combineByKey()
- aggregateByKey()

Phase 5: Partitioning & Performance

- Analyze number of partitions
- Use `cache()` or `persist()`
- Compare execution time before/after caching
- Use `repartition()` or `coalesce()`
- Discuss performance observations